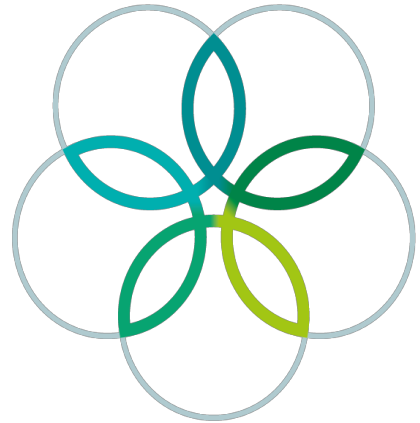
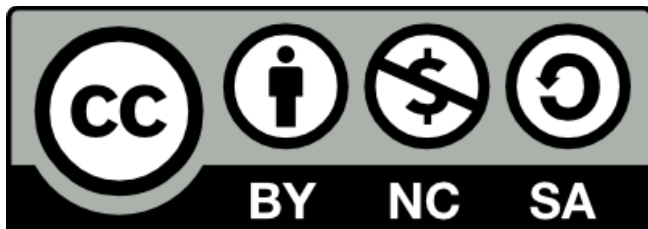


INTERNATIONAL
BIOLOGY
OLYMPIAD e. V.

IBO



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4. 3. Theoretical Test

General remarks:

This test consists of two parts, **A and B**. In part A there are 103 multiple choice questions, each having only one correct answer. In part B there are 46 questions, each of which may have more than one answer.

In order to eliminate the consequence of guessing in the marking of Type A questions, one point will be deducted for every 5 incorrect answers. Failing to answer a question will not result in any penalty.

In the marking of Type B questions a percentage of the total mark for that question will be deducted for each incorrect answer. The minimum mark for each question with deductions will be zero.

For the multiple choice questions (type A) mark the correct answer with a cross “X” in the blank space provided. If you want to change your answer with a new one in order to cancel it, you may draw a parallel line on your old mark.

correct

X

delete

~~X~~

4.3.1. Theoretical Test - Part A

CELL BIOLOGY

1. In which way are the proteins transported from the site of synthesis to the cell membrane for secretion?

- ☐ A) By cytoplasmic movement
- ☐ B) By some signal proteins in the cytosol
- ☐ C) By protein-carbohydrate complex carrying signals in the cytosol
- ☐ D) By cytoskeleton elements
- ☐ E) By vesicles

2. What is the major difference between a vacuole and a vesicle?

- ☐ A) The membrane is thick in the vacuole but thin in the vesicle
- ☐ B) The vesicle is pinched off only from the cell membrane; the vacuole is pinched off from the Golgi apparatus
- ☐ C) The vacuole membrane is carbohydrate rich; the vesicle membrane is protein rich
- ☐ D) The vacuole is near the nucleus; the vesicle is near the Golgi apparatus
- ☐ E) The vacuole has a comparatively slow movement; the vesicle moves rapidly

3. Which of the following is not a function of the Golgi apparatus?

- ☐ A) Addition of sugars to proteins
- ☐ B) Storage of lipids
- ☐ C) Package of secretion products
- ☐ D) Formation of glycolipids
- ☐ E) Synthesis of polysaccharides from simple sugars

4. Which of the following functions are carried out in the smooth endoplasmic reticulum?

- I) Addition of carbohydrates to proteins**
- II) Synthesis of membrane phospholipids**
- III) Addition of carbohydrates to lipids**
- IV) Synthesis of cholesterol**
- V) Detoxification of drugs**

- ☐ A) I, II, IV
- ☐ B) II, III, IV
- ☐ C) II, IV, V
- ☐ D) I, IV, V
- ☐ E) I, II, V

5. Deleted

6. Fibroblasts are cells which synthesize proteins of the extracellular matrix of the connective tissue (collagen fibers), glycoproteins (fibronectin) and proteoglycans (dermatan sulphate). According to these features which organelle/organelles has/have the greater function in these cells?

- ☐ A) Rough endoplasmic reticulum and smooth endoplasmic reticulum
- ☐ B) Golgi apparatus
- ☐ C) Rough endoplasmic reticulum and free ribosomes
- ☐ D) Golgi and rough endoplasmic reticulum
- ☐ E) Rough endoplasmic reticulum

7. Four structures (I-IV) and some related functional and structural features (1-7) are given below.

I. Cilia

II. Basal bodies

III. Centrosome

IV. Flagellum

1- There is a 9+2 arrangement in a ring around a pair of single microtubules

2- Most of these structures are longer than the cell

3- It is shorter than the cell

4- There are nine groups of three microtubules, fused into triplets with an empty core

5- These are the main structures for movement

6- They function in the synthesis of spindle fibers

7- They bind cilia and flagella to the cell membrane

Which of the combinations below for structure and function are correct?

___A) I: 1, 3, 5 II: 3, 4, 5 III: 3, 4, 6 IV: 1, 2, 3

___B) I:1, 4, 5 II: 1, 2, 7 III: 2, 3, 4 IV: 1, 3, 5

___C) I:1, 4, 7 II: 3, 4, 5 III: 2, 3, 6 IV: 2, 3, 4

___D) I:3, 4, 6 II: 2, 4, 7 III: 3, 4, 5 IV: 4, 5, 6

___E) I:2, 4, 6 II: 2, 4, 7 III: 3, 4, 5 IV: 2, 4, 5

8. Which of the following contains a polar head and a non-polar tail in cell?

___A) Triglycerides ___B) Neutral lipids ___C) Wax

___D) Phospholipids ___E) All the above

9. Which of the following are the fibers that attach to the the cytoplasmic face of spot desmosomes?

- ___A) Collagen fibers
- ___B) Cytoskeleton fibers
- ___C) Elastic fibers
- ___D) Tubulin protein fibers
- ___E) Reticular fibers

10. Which of the following is the correct description of a microsome?

- ___A) It consists of vesicles detached from the Golgi apparatus
- ___B) It consists of vesicles containing waste products digested by the lysosomes
- ___C) It consists of various amounts of ribosomes and fragmented endoplasmic reticulum
- ___D) It is a vacuole that contains secretions
- ___E) It is a ribosome dimer

11. In aerobic respiration glucose is converted to pyruvate in the

- ___A) Inner mitochondrial membrane ___B) Cytoplasm
- ___C) Outer mitochondrial membrane ___D) Mitochondrial matrix
- ___E) Mitochondrial membrane interspace (intermembrane space)

12. Which of the following is the correct description for a porin?

- ___A) It is a protein in the structure of microtubules
- ___B) It is a protein located on the outer mitochondrial membrane
- ___C) It is a protein of the nuclear pores
- ___D) It is a lipid that functions in the addition of carbohydrates to proteins
- ___E) It is a protein that forms the cytoskeleton

13. How are the peroxisomes formed in a cell?

- ☐ A) Only by fission
- ☐ B) Only by detachment from a big peroxisome
- ☐ C) Both by fission and self-replication of a preexisting peroxisome
- ☐ D) Only by budding from plasma membrane
- ☐ E) Only by self-replication

14. In living cells there are

- | | |
|----------------------------|--------------------------|
| 1. Ribosomes | 5. Introns |
| 2. ATP synthesis | 6. DNA polymerase |
| 3. Cell membrane | 7. Photosynthesis |
| 4. Nuclear envelope | 8. Mitochondria |

Which of them can exist both in prokaryotic and eukaryotic cells?

- ☐ A) 1, 2, 3, 6, 7 ☐ B) 1, 2, 3, 5, 7, 8 ☐ C) 1, 2, 3, 4, 7
- ☐ D) 1, 3, 5, 6 ☐ E) 2, 3, 7, 8

15. Specific inhibitor "X" of F_0F_1 ATPase is added to a rat liver cell carrying out the oxidation of glucose under aerobic conditions. Which of the following would not occur as a result of this inhibition?

- ☐ A) Mitochondrial ATP formation will stop
- ☐ B) The citric acid cycle will slow down because of insufficient NAD^+ regeneration
- ☐ C) The rate of glucose consumption will decrease
- ☐ D) Glycolysis will be accelerated
- ☐ E) Oxygen consumption will be halted

16. Agarose gel electrophoresis was applied to the DNA samples given below. What will be the order of migration from the well, at the completion of the electrophoresis?

I- F⁺ bacterial plasmid

II- F' bacterial plasmid

III- Hfr *E. coli* chromosomal DNA

IV- *E. coli* chromosomal DNA

- ___A) I, II, III, IV ___B) II, III, I, IV ___C) IV, III, II, I
___D) III, IV, II, I ___E) IV, I, III, II

17. If you observe two DNA samples X and Y (each containing 1200 base pairs) migrating at different rates in an agarose gel, what would your interpretation be ?

- ___A) The amount of adenines in sample X is greater
___B) The amount of guanines in sample Y is greater
___C) The percentage of agarose in the gel is greater than 0.8 %
___D) There are intercalating agents in the agarose gel
___E) Samples X and Y have different conformations

18. The enzyme phosphofructokinase;

I- It is the major regulatory enzyme in glycolysis

II- ATP is the substrate for the enzyme

III- ATP is the negative modulator of the enzyme

IV- Citrate activates the enzyme

Which of the following is the correct answer for the above statements concerning the enzyme phosphofructokinase?

- ___A) Only IV is correct ___B) Only I and III are correct
___C) Only I, II and III are correct ___D) Only II and IV are correct
___E) I, II, III and IV are correct

19. If oligomycin and 2,4-dinitrophenol are both added;to a suspension of mitochondria containing substrates, P_i (inorganic phosphate), Mg^{++} and ADP;

- ☐ A) Both O_2 consumption and ADP phosphorylation will cease
- ☐ B) The rate of O_2 consumption will increase but ADP phosphorylation will cease
- ☐ C) Phosphorylation/ O_2 consumption ratio will remain the same
- ☐ D) Phosphorylation/ O_2 consumption ratio will increase
- ☐ E) O_2 consumption will decrease but ADP phosphorylation will continue

20. In anaerobic glycolysis 2 moles of inorganic phosphate (P_i) are used for one mole of glucose consumed. Which of the following enzymes catalyzes the reaction in which P_i is directly consumed?

- ☐ A) Hexokinase
- ☐ B) Phosphofructokinase
- ☐ C) Pyruvate kinase
- ☐ D) Glyceraldehyde-3-phosphate dehydrogenase
- ☐ E) Enolase

21. Which of the following cannot use ketone bodies for the generation of energy?

- ☐ A) The brain (in fasting)
- ☐ B) The heart muscle
- ☐ C) Erythrocytes
- ☐ D) The kidney cortex
- ☐ E) The skeletal muscle

Answer Key-Theoretical Test-Part A

22. Inside the chloroplast the potential uses for the G3P (glyceraldehyde 3-phosphate) produced in the Calvin cycle include the synthesis of:

- ☐ A) Fatty acids
- ☐ B) Glycerol
- ☐ C) Glucose
- ☐ D) Amino acids
- ☐ E) All of the above

23. The following statements are about the effect of a competitive inhibitor in a reaction catalyzed by an enzyme.

- I- $V_{\max}$ is unchanged**
- II- The inhibition can be reversed by increasing the concentration of the substrate**
- III- K_m increases**
- IV- The inhibitor binds to the enzyme at a different site than the active site**

Which combination of statement(s) is/are true?

- ☐ A) I, II and III ☐ B) Only I and III ☐ C) Only II and IV
- ☐ D) Only IV ☐ E) I, II, III and IV

24. Which of the following bonds is not present in the structure of DNA?

- ☐ A) 3'-5' phosphodiester bond
- ☐ B) N-glycosidic bond
- ☐ C) H-bonds
- ☐ D) Hydrophobic interactions
- ☐ E) Disulphide bonds

25. Consider these two relationships and the four statements about the aminoacids/proteins and fatty acids/triglycerides

Left	Right	Left	Right
Amino acids:	Proteins	Fatty acids:	Triglycerides

- I. Both molecules on the right consist only of repeated units of the molecules on the left
- II. In the process of synthesis of both molecules on the right, at least some electrical charges are neutralized
- III. In both relationships, the diversity in the molecules on the left result in the diversity in the molecules on the right
- IV. In the synthesis of both molecules on the right, water is released

Which of the statement(s) is/are correct?

- ___A) I, II, III, and IV ___B) II, III and IV ___C) III and IV
 ___D) Only III ___E) Only IV

26. Five different cell cultures were treated with different radioactively labelled compounds as follows:

<u>Compound</u>	<u>Cell Culture</u>
Lactose	Cell culture a
Valine	Cell culture b
Thymidine triphosphate	Cell culture c
Glutamic acid	Cell culture d
Alanine	Cell culture e

After an hour the cells were washed, fixed and autoradiographed. In order to study nuclear activities *in vivo* which of the cell cultures is the best?

- ___A) Cell culture a ___B) Cell culture b ___C) Cell culture c
 ___D) Cell culture d ___E) Cell culture e

27. Which of the molecules is responsible for the autocatalytic excision of introns and splicing of exons in eukaryotic cells?

- ☐ A) RNA polymerase
- ☐ B) Ribonuclease
- ☐ C) Ribozyme
- ☐ D) Reverse transcriptase
- ☐ E) Endonuclease

28. The interaction between the anticodon of a tRNA molecule and the complementary codon of mRNA is achieved by:

- ☐ A) The catalysis by peptidyl transferase
- ☐ B) ATP energy
- ☐ C) The catalysis by amino-acyl-tRNA synthetase
- ☐ D) The covalent bonds formed with energy from GTP
- ☐ E) H-bonds

29. The lac (lactose) operon is an example of:

- ☐ A) Translational control
- ☐ B) Posttranslational control
- ☐ C) Replication control
- ☐ D) Transcriptional control
- ☐ E) All of the above

30. The breakdown of glucose in the cell is controlled by the activation or inactivation of enzymes present in the specific stages of glycolysis and the citric acid cycle. There are three key enzymes like this. The conditions that activate or inactivate these enzymes are given in the table below. Which combination is correct for the activation of all the three enzymes?

	ENZYMES		
	Phosphofructokinase	Citrate synthase	Isocitrate dehydrogenase
___A)	High level of ADP+ AMP	Low level of ATP+NADH	Low level of ATP+NADH
___B)	High level of ATP	High level of ATP+NADH	Low level of ATP+NADH
___C)	Low level of ATP	High level of ATP+NADH	Low level of ATP+NADH
___D)	High level of ADP andAMP	High level of ATP+NADH	High level of eitherADP or NAD ⁺
___E)	High level of ATP	Low level of either ATP or NADH	High level of either ATP or NAD ⁺

31. Which of the following statements is false about prokaryotic RNA polymerase?

- ___A) The synthesis is in the 5'-3' direction
- ___B) There is only one RNA polymerase enzyme responsible for the synthesis of rRNA, m-RNA and t-RNA
- ___C) Its RNA product will hybridize with the DNA template
- ___D) The transcription starts from the AUG codon in the DNA
- ___E) The enzyme synthesizes a single transcript that codes for several polypeptide chains

Answer Key-Theoretical Test-Part A

32. Which of the following statement about the regulator gene in bacterial operon model is true?

- ☐ A) It codes for repressor protein
- ☐ B) It codes for inducer molecules
- ☐ C) It is the binding site of RNA polymerase
- ☐ D) It is the binding site of inducer molecules
- ☐ E) Provides the transcription or inhibition of transcription of the structural genes

33. Which of the following statements is false regarding to the procaryotic mRNA?

- ☐ A) It is polycistronic
- ☐ B) It does not involve introns
- ☐ C) It binds to ribosome from the 5'end.
- ☐ D) It is synthesized in the nucleus
- ☐ E) It can form a single transcript that codes for several polypeptides.

34. Which of the following statements is a false description for a codon?

- ☐ A) It consists of three nucleotides
- ☐ B) It is the basic unit of the genetic code
- ☐ C) There may be more than one codon for the same amino acid
- ☐ D) It is located on the t-RNA
- ☐ E) It can never code more than one amino acid

GENETICS AND EVOLUTION

35. A cross was made between two albinos and phenotypically identical F_1 generation was obtained. When F_1 was self-crossed, F_2 was observed as 9 normal and 7 albinos. Which of the following combinations suits this kind of inheritance?

	Parents	O f f s p r i n g s (F_2)			
___A)	AAbb X aaBB	9A-B-	3aaB-	3A-bb	1aabb
___B)	aabb X AAbb	9A-B-	3aaBb	3Aabb	1aabb
___C)	AaBb X AaBb	9A-B-	3aaBb	3Aabb	1aabb
___D)	aaBb X Aabb	9A-B-	3aaB-	3A-bb	1aabb
___E)	AABB X aabb	9A-B-	3aaB-	3Aabb	1aabb

36. In an experimental population, the frequency of the O blood type is 25%, A is 24%, B is 39% and AB is 12%. Which of the following is the frequency of the alleles which are responsible for the blood types A, B and O?

	<u>A</u>	<u>B</u>	<u>O</u>
___A)	0.3	0.2	0.5
___B)	0.2	0.5	0.3
___C)	0.2	0.3	0.5
___D)	0.5	0.2	0.3
___E)	0.3	0.5	0.2

37. Deleted

38. Incomplete penetrance, sex-limited traits, sex-influenced traits, age-influenced traits and temperature-influenced traits are all examples of

- ___A) Linkage ___B) Conditional gene expression
___C) Epistasis ___D) Multiple alleles ___E) Partial dominance

39. Deleted

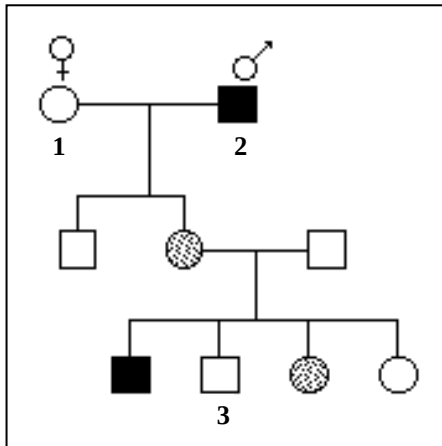
40. The fruit weights of a squash plant vary between 2 and 4 kg. The fruit weights are a product of pairs of additive polygenic genes. Which of the following is true for the F₂ generation that resulted from a 2 kg squash being pollinated with a 4 kg one in terms of the number of individuals of any weight classes and also regarding the genotypes of the pollinated 2 kg and 4 kg squashes.

	4 kg	3.5 kg	3 kg	2.5 kg	2 kg	2 kg parent	4 kg parent
___A)	1	2	6	2	1	aabb	AABB
___B)	1	4	6	4	1	AaBb	aaBB
___C)	1	4	6	4	1	AAbb	aaBB
___D)	1	6	2	6	1	AABb	AABB
___E)	1	4	6	4	1	aabb	AABB

41. Which of the following is not a reason why recessive alleles are not observed in the phenotypes of heterozygotes?

- ___A) The recessive allele codes for a nonfunctioning protein
___B) The recessive allele is linked to the dominant allele
___C) The dominant allele produces so much product as to “swamp” the product of the recessive allele
___D) The recessive allele is normal, but the product of the dominant allele inhibits the function of the recessive allele
___E) The allele’s product (e.g. an enzyme) is much less functional and therefore masked by the dominant allele’s product

42.



Healthy : Empty

Sick : Dark colored

Carrier : Spotted

On the pedigree (family tree) given above which of the following would be the genotype of the individuals written as 1, 2 and 3?

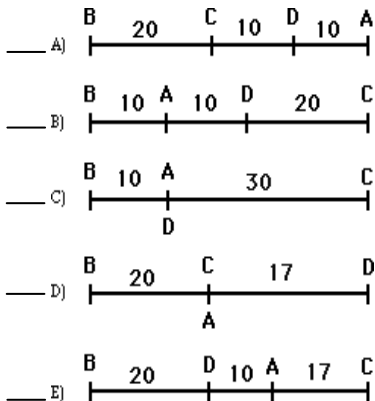
	1	2	3
___A)	AA XX	a XY	A XY
___B)	Aa	Aa	aa
___C)	aa XX	A XY	A XY
___D)	aa	Aa	aa
___E)	Aa	AA	aa

Answer Key-Theoretical Test-Part A

43. An individual of the genotype AaBbCcDd was crossed with the one aabbccdd and the following results were obtained.

aBCD	42
Abcd	43
ABCd	140
abcD	145
aBcD	6
AbCd	9
ABcd	305
abCD	310

Which of the following shows the arrangement of the genes and their distance (in centimorgans)?

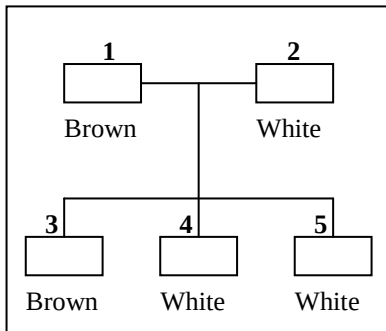


44. Genetic variations are important for populations

- ___A) So that males and females of parthenogenetic species might be distinguished
- ___B) So that evolution is directed
- ___C) Because they provide the raw material on which selection acts
- ___D) So that organisms might be classified
- ___E) To make them more interesting to study

45. When a dominant allele (A) is alone, it causes a brown fur color but when it is with another allele which has an epistatic effect, the fur color is white.

According to this, which of the following shows the true genotypes of the individuals in the family tree given above?



- | | 1 | 2 | 3 | 4 | 5 |
|-------|------|------|------|------|------|
| ___A) | Aabb | AaBb | Aabb | AaBb | aabb |
| ___B) | AaBb | aabb | AaBb | aaBb | aabb |
| ___C) | AaBb | aaBb | AaBb | Aabb | Aabb |
| ___D) | AaBb | aabb | AaBb | aaBb | aabb |
| ___E) | aaBb | AaBb | Aabb | aaBb | aabb |

46. Deleted

47.

	Phenotypes of the individuals	
Genotypes	Female	Male
AA	Δ	Δ
Aa	Δ	Δ
aa	Δ	$\emptyset$

An inheritance scheme of any character is given above. Which of the following is true for the inheritance type of this character?

- ___A) This character is sex linked
- ___B) This character is sex limited
- ___C) This character is sex influenced
- ___D) Incomplete penetrance is seen in the inheritance of this character
- ___E) Codominance is seen in the inheritance of this character

48. In cattle, the polled (hornless) condition is dominant to the horned condition. Coat colour can be red, white or roan (red with white patches). Both genes are carried on autosomes and they are not linked. A cross was carried out between a cow and a bull, both of which had the roan coat colour and both were heterozygous for the polled condition. Which of the following statements are true about the offspring from the cross, assuming that the cross was carried out several times to produce a lot of offspring?

- 1. The chance of producing white polled and white horned offspring is the same.**
- 2. The chance of producing roan polled offspring is three times that of producing roan horned.**
- 3. There is an equal chance of producing red polled and white polled offspring.**
- 4. Statistically there should be more roan horned offspring than any other type.**
- 5. The chance of producing roan polled offspring is twice that of producing white polled.**

___A) 1 & 2

___B) 2 & 3

___C) 3 & 4

___D) 1, 2 & 3

___E) 2, 3 & 5

Answer Key-Theoretical Test-Part A

49. In guinea-pigs, there are several alleles involved in determining the animal's coat color. C^b - black; C^c - creamy; C^s - silver and C^z - albino. Analyze the results of the following crosses and determine the most suitable order of alleles referring to dominance-recessiveness relationships of these alleles.

Crosses	Phenotype of parents	Phenotype of offspring			
		Black	Silver	Creamy	Albino
1	black x black	22	0	0	7
2	black x albino	10	9	0	0
3	creamy x creamy	0	0	30	11
4	silver x creamy	0	23	11	12

___A) $C^b > C^c > C^s > C^z$

___B) $C^b > C^s > C^c > C^z$

___C) $C^c > C^z > C^b > C^s$

___D) $C^b > C^z > C^s > C^c$

___E) $C^b > C^c > C^z > C^s$

50. Deleted

51. Which of the following is not a proof that eukaryotic cells evolved by endosymbiosis?

- ☐ A) Similarity between spirochetes and flagellum
- ☐ B) Similarity between mitochondrial DNA and procaryotic DNA
- ☐ C) Similarity between bacterial and chloroplastic ribosomes
- ☐ D) Similarity between chloroplast and cyanobacteria
- ☐ E) Similarity of the inhibition of the protein synthesis between the eucaryotic cells and the mitochondria

52. Which of the following has the best evolutionary adaptation capacity?

- ☐ A) Primitive, heterogenous heredity material, high number of generation, short life span
- ☐ B) Highly specific, homogenous heredity material , high number of generation, long life span
- ☐ C) Highly specific feeding regime, living underground, asexual reproduction
- ☐ D) Living on high mountains, nocturnal, feeding on the most common plants
- ☐ E) Highly tolerant to hereditary changes, low offspring success, specific feeding regime

53. Which of the following is not a biological characteristic of desert organisms?

- ☐ A) No regular reproductive cycle
- ☐ B) Seeds germinate immediately after flowering and fruiting
- ☐ C) Fewer stoma (in plant)
- ☐ D) Specialized kidney capable of reabsorbing water (in animal)
- ☐ E) More succulent plants

54. Deleted

55. Biologists assume that the first heredity material to appear was RNA. Which of the following may be the main reason for that?

- ☐ A) RNA was produced in Miller's experiment
- ☐ B) RNA is structurally more primitive than DNA
- ☐ C) The RNA called ribozyme catalyses some chemical reactions
- ☐ D) DNA can not stay stable in hydrophobic medium
- ☐ E) RNA appears in all animals

56. Which of the following is not evidence that higher plants are derived from green algae?

- ☐ A) Some green algae have multicellular sporophyte and gametophyte phases
- ☐ B) Both plants and algae have cellulose in their cell walls
- ☐ C) Both plants and algae have similar photosynthetic and accessory pigments
- ☐ D) Both plants and algae synthesize starch as a main store product
- ☐ E) Green algae and higher plants have the same amount of DNA per cell

57. The statements below are about various pollination strategies in plants. Which one could be the most disadvantageous for evolution of new species?

- ☐ A) The stigma can recognize the origin of pollen grains, and does not accept those from the same flower
- ☐ B) The stigma never emerges from the corolla, and only accepts pollen grains from the same flower
- ☐ C) The corolla forms a long tube, only allowing some specialized pollinators which carry pollen grains from the same species to enter
- ☐ D) The stamens and pistil mature at different times
- ☐ E) The stamens and pistil are located in different flowers

58. Which of the following triplets is false for the hearing organ?

	<u>Structure</u>	<u>Animal</u>	<u>Function</u>
___A)	Columella	Lizard	Transports the sound from the membrane to the cochlea
___B)	Weber bones	Fish	Transports the sound created by the vibration of the swim bladder to the brain
___C)	Tarsal bones	Salamander	Transports the vibrations from the soil to the inner ear
___D)	Some cranial	Whale	Transports the sound from water to the (otic) bones inner ear
___E)	Utricle (sacculle)	Mole	Transports the vibrations from the soil to the inner ear

59. Deleted

PLANT ANATOMY AND PHYSIOLOGY

60. Which of the following can not be stated relating to cyclic photophosphorylation?

Note: $\text{NADP}_{\text{red}} = \text{NADPH}$; $\text{NADP}_{\text{ox}} = \text{NADP}^+$

- ___A) It is favored when the cell is more in need of ATP than NADP_{red}
- ___B) It is favored when NADP_{ox} is in short supply
- ___C) An energized electron is first accepted by ferredoxin
- ___D) Plastocyanin is the last acceptor of an energized electron before it reaches the center
- ___E) In the system, cytochrome “f” connects ferredoxin to plastoquinone

61. Deleted

62.

- I. It is required for activity of some dehydrogenases, decarboxylases, kinases, oxidases and peroxidases**
- II. Under its deficiency, plant tissues become soft and often flaccid even under low temperature and stress conditions**
- III. It is required for the photosynthetic reactions involved in the O_2 cycle**

Which of the following gives the best match of minerals to the statements above?

	<u>I</u>	<u>II</u>	<u>III</u>
___A)	N	Ca	Mg
___B)	S	Mn	Mg
___C)	Mn	N	P
___D)	Mn	Ca	Cl
___E)	Cl	K	P

63. When the temperature is high and the amount of dissolved oxygen is higher than that of CO₂ in the chloroplasts, in which of the following plants does growth not slow down?

- ☐ A) Wheat
- ☐ B) Watermelon
- ☐ C) Sunflower
- ☐ D) Sugar cane
- ☐ E) Rice

64. Deleted

65. Deleted

66. Deleted

67. Deleted

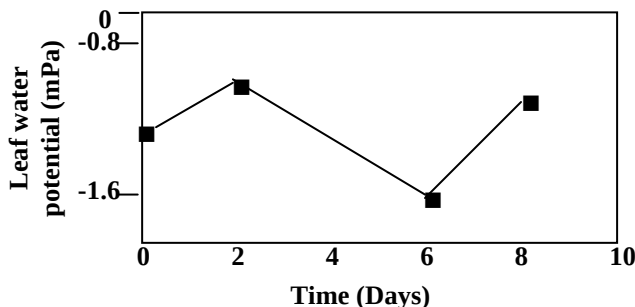
68. Which of the following cannot be referred to as blue-light responses in higher plants and fungi?

- ☐ A) Phototropism in *Phycomyces*
- ☐ B) Carotenoid biosynthesis in *Neurospora*
- ☐ C) Stomatal opening in higher plants
- ☐ D) Chloroplast rearrangements
- ☐ E) Flavenoid synthesis

69. “Shooty” tumors are produced in the stem of an “X” plant infected by bacteria in which mutations occur in their T-DNA, while “rooty” tumors are formed in the stem of a “Y” plant. Which of the following is true for the effects mentioned above.

- ☐ A) In the “X” plant, the genes which are responsible for giberellin (giberellic acid) synthesis are inactivated while in the “Y” plant, genes which are responsible for abscisic acid synthesis are inactivated.
- ☐ B) In the “X” plant, the genes which are responsible for Indol acetic-acid synthesis are inactivated while in the “Y” plant, genes which are responsible for zeatin synthesis are inactivated.
- ☐ C) In the “X” plant, the genes which are responsible for zeatin synthesis are inactivated while in the “Y” plant, genes which are responsible for ethylene synthesis are inactivated.
- ☐ D) In the “X” plant, the genes which are responsible for abscisic acid synthesis are inactivated while in the “Y” plant, genes which are responsible ethylene synthesis are inactivated.
- ☐ E) In the “X” plant, the genes which are responsible for cytokinins synthesis are inactivated while in the “Y” plant, genes which are responsible for ethylene synthesis are inactivated.

70. In the figure, changes in water potential in the leaves of a plant over a period of time are shown. Which of the following is true according to this situation?



- ___A) The ABA (Abscissic acid) content decreased, and stomata resistance increased between the 2nd and 6th days; the process was reversed between the 6th and 8th days
- ___B) The ABA content did not change and stomata resistance decreased between the 2nd and 6th days; the process was reversed between the 6th and 8th days
- ___C) The ABA content increased and stomata resistance decreased between the 2nd and 6th days; the process was reversed between the 6th and 8th days
- ___D) The ABA content and stomata resistance increased between the 2nd and 6th days; the process was reversed between the 6th and 8th days.
- ___E) The ABA content decreased and stomata conductance increased between the 2nd and 6th days; the process was reversed between the 6th and 8th days.

71. Which of the following are limiting or near-limiting nutrients both in aquatic and terrestrial systems?

- ☐ A) Nitrogen-potassium ☐ B) Potassium-magnesium
- ☐ C) Phosphorus-nitrogen ☐ D) Calcium-magnesium
- ☐ E) Iodine-magnesium

72. Which of the following is not true for the auxin transport in plants?

- ☐ A) IAA transport usually does not take place in sieve tubes and xylem
- ☐ B) IAA transport usually occurs in parenchymatic cells adjacent to vascular bundles
- ☐ C) Auxin moves rather slowly within the plant
- ☐ D) IAA moves mainly from the apex to the base (basipetal direction)
- ☐ E) Auxin transport does not require energy

73. Light is perceived by all living organisms in one way or the other. The pigment which is chosen for this process of photoperception are carotenoids. Which of the following properties make carotenoids the right pigment for this function.

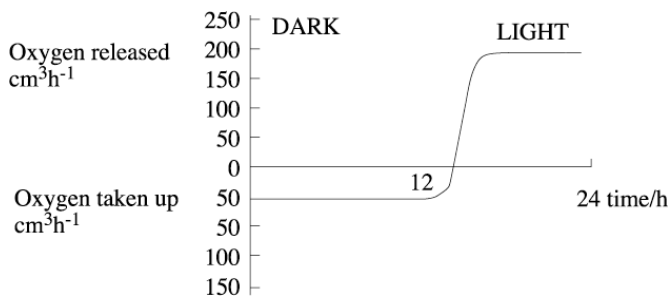
- ☐ A) Their ability to absorb most of the visible and ultraviolet light
- ☐ B) Their high capacity to store and transfer light energy as chemical energy
- ☐ C) As saturated organic compounds, their capability to preserve themselves against environmental factors such as high energy currents
- ☐ D) Their high affinity for proteins which have a role in perception
- ☐ E) The efficiency of long structures of alternating double bonds in their structure to initiate light sensitive stereoisomerism

74. The changes that take place in climacteric fruits when they ripen (colour, texture and chemical composition) are mainly due to:

- ___A) The CO₂ content in the atmosphere
- ___B) The temperature variation
- ___C) The ethylene synthesis in the fruit
- ___D) The auxin concentration in the fruit
- ___E) Gibberellin concentration in fruit

75. Deleted

76. The data were obtained relating to the rates of oxygen release and uptake in plants. The plants were placed in the dark for 12 hours followed by 12 hours in light. The temperature was constant throughout the experiment. The results are shown in the graph.



Which of the following is the most accurate estimate of the total volume of oxygen used by the plants for respiration during 24 hours of the experimental time?

- ___A) 50 cm³
- ___B) 600 cm³
- ___C) 1000 cm³
- ___D) 1200 cm³
- ___E) 1800 cm³

Answer Key-Theoretical Test-Part A

77. In which aspect does C_4 photosynthesis differ from Crassulacean acid metabolism (CAM)?

- ☐ A) PEP carboxylase is only used in C_4 photosynthesis
- ☐ B) CO_2 fixation in CAM plants occurs at night while it occurs in C_4 plants during the day time
- ☐ C) Organic acids with four carbons are only produced in C_4 photosynthesis
- ☐ D) Only plants with crassulacean acid metabolism can carry out photosynthesis in arid environments
- ☐ E) Only plants with C_4 photosynthesis can economize water

78. Which of the following cannot be stated with relation to the shoot apex?

- ☐ A) There is only one apical cell in vascular non-flowering plants
- ☐ B) There is more than one apical cell in each cell layer in gymnosperms
- ☐ C) There are different apical cells in more than one tissue layer in angiosperms
- ☐ D) An apical cell is pyramidal shaped in non-flowering plants
- ☐ E) A shoot apex with a distinct tunica and corpus is found in gymnosperms

79. Which of the following cannot be stated for the collenchyma?

- ☐ A) It is a living tissue found in developing organs
- ☐ B) It is formed in the roots only under the effect of light
- ☐ C) Its location in petioles is peripheral
- ☐ D) It is located at the periphery of woody stems
- ☐ E) It is located at the periphery of lamina

80. Which of the following cannot be referred to as a function of the sporoderm (exine) layer of pollens?

- ___A) The storage of enzyme proteins for the reactions
- ___B) Playing a role in the reaction between pollen and stigma
- ___C) Production of the pollen tube
- ___D) Protection of pollen against external factors
- ___E) Realizing pollination

81. Deleted

82. In a flower, flower symmetry is radial, calyx 4 and fused, corolla 4 and is separate, the androecium has 5 stamens and is connected to the corolla, the gynoecium is compound in 5 parts, superior and syncarpous. According to the description given above, which of the formulae in the following is correct?

- ___A) $+ K_{(4)} [C_4A_{(5)}] \underline{G_{(5)}}$
- ___B) $+ K_{(4)} C_4A_{(5)} \underline{G_{(5)}}$
- ___C) $*K_4 C_4A_5 \underline{G_5}$
- ___D) $*K_{(4)} [C_4A_{(5)}] \underline{G_{(5)}}$
- ___E) $*K_4 [C_4A_5] \underline{G_5}$

83. Which of the following is true for a C_4 plant in which some leaves can carry out C_3 photosynthesis while others can carry out C_4 photosynthesis?

- ___A) In fact, it is a C_3 plant
- ___B) The leaves which carry out C_3 photosynthesis lack Kranz anatomy
- ___C) PEP (phosphoenolpyruvate) is not synthesized in the leaves which carry out C_4 photosynthesis
- ___D) It indicates that the C_4 pathway was evolved from the C_3 pathway
- ___E) Both C_3 and C_4 photosynthesis do not occur on the same leaf

84. Deleted

BIOSYSTEMATICS

85. Deleted

86.

- I. A small dry single-seeded, indehiscent fruit**
- II. A fruit with a single ovary consisting of a single carpel**
- III. Dehiscent fruit is formed by two carpels with a septum between the carpels and its length is less than three times of its width**

The above statements describe three different fruit types.

I

II

III

Which of the following combinations are correct for the fruit types?

	<u>I</u>	<u>II</u>	<u>II</u>
___A)	Siliquea	Legume	Achene
___B)	Legume	Siliquea	Achene
___C)	Siliquea	Achene	Legume
___D)	Achene	Siliquea	Legume
___E)	Achene	Legume	Siliquea

87. Deleted

88. Deleted

89. Which one of the following is not a characteristic of a deuterostomian animal ?

- ☐ A) Radial cleavage during the embryonic development
- ☐ B) Regulative development during the embryonic period
- ☐ C) Enterocoelom
- ☐ D) Pharyngeal slits on the pharynx
- ☐ E) Original (evolutionary origin) bilateral symmetry

90. The main reason for echinoderms living only in the sea is because;

- ☐ A) They were adapted to be sessile so they do not have a great distribution
- ☐ B) They appear first in the seas
- ☐ C) They live in different habitat types in the sea
- ☐ D) They have no excretory system
- ☐ E) There are safer places in the deep sea

91. Which one of the following structures of earthworms has similar functions to the liver of vertebrates?

- ☐ A) Typhlosolis
- ☐ B) Coelomocytes
- ☐ C) Chloragogen cells
- ☐ D) Cells that line the inner surface of the small intestine
- ☐ E) Calcium gland cells

92. The metanephridia of annelids and molluscs are functionally and structurally similar to the vertebrate kidneys. During the formation of urine, filtration, reabsorption and secretion processes occur. Where does filtration occur in the nephridium of mussels?

- ☐ A) On the nephrostom in the nephridium
- ☐ B) On the cardiac wall and pericardial glands
- ☐ C) On the tubules that are connected to the nephrostom
- ☐ D) On the wall of the intestine
- ☐ E) On the gill capillaries

93. Which one of the following is not a characteristic of molluscs (Mollusca)?

- ☐ A) Mantle
- ☐ B) Radula
- ☐ C) Trochophore larva
- ☐ D) Spiral cleavage
- ☐ E) Regulative development

94. Deleted

ANIMAL ANATOMY AND PHYSIOLOGY

95. Deleted

96. Which of the following is an important feature of primitive aquatic life forms?

- ___A) Partially oxygenated blood
- ___B) An open circulation with no small blood vessels or capillaries
- ___C) Significantly decreased blood pressure
- ___D) Highly acidic blood
- ___E) Carriage of most of the O₂ in the the plasma

97.

I- Partial O₂ pressure

II- pH

III- Amount of 2,3-diphosphoglycerate

IV- Partial CO₂ pressure

V- Body temperature

The factors affecting the dissociation of O₂ from hemoglobin in the human circulatory system are given above. In which of the following alternatives does this dissociation occur most easily?

- | | | |
|--------------------|---------------|---------------|
| ___A) I increases | III decreases | IV increases |
| ___B) II increases | V decreases | III increases |
| ___C) I decreases | IV increases | III increases |
| ___D) V increases | IV decreases | I increases |
| ___E) II decreases | III decreases | V decreases |

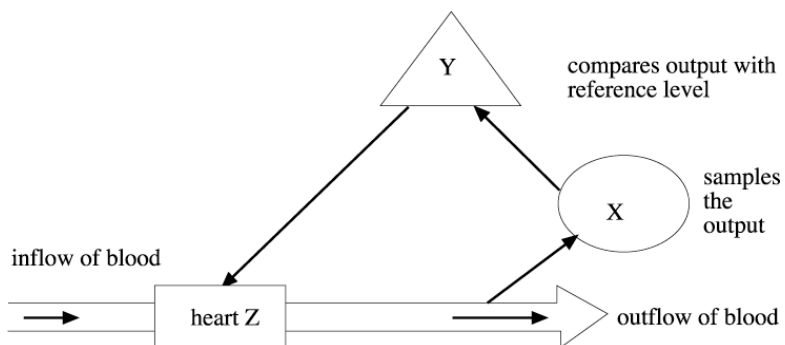
Answer Key-Theoretical Test-Part A

98. A nerve-skeletal muscle isolated preparation is placed in a Ca^{2+} - free medium appropriate for its survival. If the nerve is electrically stimulated, which of the following events will occur?

- ☐ A) The muscle will not be stimulated
- ☐ B) The muscle will be stimulated but will not contract
- ☐ C) The muscle will be both stimulated and contract
- ☐ D) The muscle will not be stimulated, but even if it is, it will not contract
- ☐ E) The muscle may be stimulated, and may contract but it will not relax

99. Which of the following is false about the differences between the vertebrate skeletal muscles and smooth muscles?

- ☐ A) Skeletal muscle is more sensitive to electrical stimulation while smooth muscle is more sensitive to chemical stimulation
- ☐ B) Skeletal muscle has a certain length in the resting state; smooth muscle has not
- ☐ C) Smooth muscle contracts more than skeletal muscle after stretching
- ☐ D) Skeletal muscle consumes 10% less of the energy than that of smooth muscle for the same degree of contraction
- ☐ E) Without a nerve connection, skeletal muscle cannot function normally but smooth muscle can



100. The figure shows a feedback system for the control of the output of blood from the heart (cardiac output). Which of the following gives the correct description of the parts played by the components X, Y and Z?

- | | <u>X</u> | <u>Y</u> | <u>Z</u> |
|-------|----------|----------|----------|
| ___A) | Monitor | Receptor | Effector |
| ___B) | Monitor | Effector | Receptor |
| ___C) | Receptor | Monitor | Effector |
| ___D) | Receptor | Effector | Monitor |
| ___E) | Effector | Monitor | Receptor |

101.

- I- The magnitude of the impulse is dependent on the size of the stimulus**
- II- The number of fibres which are stimulated increases with the size of the stimulus**
- III- The speed at which the impulse travels increases with the size of the stimulus**
- IV- The speed at which the impulse travels depends on whether or not the nerve has a myelin sheath**
- V- The speed of the impulse conduction is directly proportional to the diameter of the axon**

Which of the following is the correct combination of the statements given above about the nerve conduction?

- ___A) I, II and III ___B) II, III and IV ___C) II, IV, and V
___D) III, IV and V ___E) I, III and V

102. Deleted

**103. The figure shows some parts of a mammalian eye numbered 1-5.
If light suddenly strikes the eye, which of the following will be the nervous pathway for the evoked unconditional pupil reflex?
(CNS= Central Nervous System)**

- ___A) From 4 to the CNS and then to 3
___B) From 1 to 4 then to the CNS and then to 2
___C) From 3 to the CNS and back to 3
___D) From 5 to 1 then to 2
___E) From 4 to the CNS and then to 5

104. When an epinephrine (adrenalin)solution is dropped on the surface of a frog muscle(*M. gastrocnemius*) *in vitro*, the muscle displays a strong contraction. However,when the epinephrine solution is injected into the muscle cell, nothing happens. Which of the following is the reason for this?

- ___A) Epinephrine induced the antagonistic effect inside the cell
- ___B) Epinephrine induced the side-effect inside the cell
- ___C) Epinephrine was not processed by proteolytic enzyme
- ___D) Epinephrine did not find the receptor inside the cell
- ___E) Epinephrine was degraded inside the cell

105. Which of the following alternatives constitute the cell groups that function effectively in the human immune system?

- ___A) T lymphocyte – B lymphocyte – Macrophage
- ___B) T lymphocyte – Macrophage – Erythrocyte
- ___C) B lymphocyte – Kupffer cell – Lipocyte
- ___D) Dendritic cell – Neutrophilic leukocyte – Fibroblast
- ___E) Microglia – Histiocyte – Megakaryocyte

106. The left-hand diagram shows a frog sciatic nerve lying across a number of electrodes. The electrodes A and B are used for stimulating and C and D for recording. The right-hand diagram shows a typical recorded action potential. Based on this information, which of the following statements is correct?

- ☐ A) The duration of the recorded action potential (d) will be independent of the distance between electrodes C and D
- ☐ B) The magnitude of the recorded action potential (m) will be independent of the distance between electrodes C and D
- ☐ C) The first deflection on the recording occurs when electrode C is negative with respect to D
- ☐ D) The duration of the recorded action potential will depend on the distance between B and C
- ☐ E) The recorded action potential can be made monophasic by applying a local anesthetic at A

107. Deleted

108. Which one of the following is incorrectly matched?

- ☐ A) Bird – Discoidal cleavage – Erythrocyte with nucleus
- ☐ B) Frog – Mesonephros kidney – Holoblastic unequal cleavage
- ☐ C) Reptile – Viviparous organisms – Telolecithal egg
- ☐ D) Fish – Deuterostomia – Radial cleavage
- ☐ E) Mussel – Protostomia – Mosaic development

109. Diagram shows the rates of filtration (F), reabsorption (R) and excretion (E) of a substance (X) in relation to its plasma concentration by the kidneys. Which one of the following statement is incorrect?

- ☐ A) The reabsorption of X is dependent on its plasma concentration
- ☐ B) The filtration rate of X is directly proportional to its plasma concentration
- ☐ C) When the plasma concentration of X reaches a certain value, its excretion rate suddenly increases
- ☐ D) The concentration of X in the urine is expected to be higher than its amount filtered in the glomerulus
- ☐ E) The filtration rate of X in the glomerulus is fixed

110. Which of the following hormonal conditions of a woman is suitable in her late pregnancy?

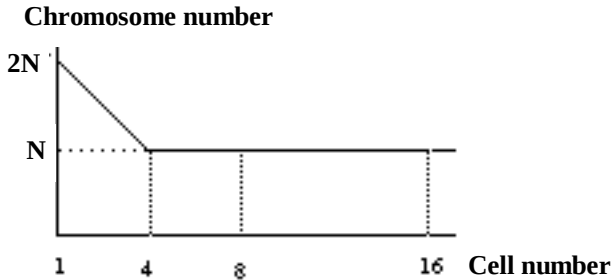
- ☐ A) Estrogen increases, progesterone increases
- ☐ B) Estrogen decreases, progesterone decreases
- ☐ C) Estrogen increases, progesterone decreases
- ☐ D) Estrogen decreases, progesterone increases
- ☐ E) Luteinizing Hormone increases, Human Chorionic Gonadotropin increases

111. Deleted

112. Deleted

113. Deleted

114.



Which of the following can be said according to the graph given above?

I Gamete formation in human

II- Gamete formation in phanerogamia

III- Gamete formation in queen honey bee and the development of the male bee

IV- Spore formation and development in ferns

___A) I and II

___B) III and IV

___C) II and IV

___D) I, II and IV

___E) I, II and III

115. Which one of the following is true about determination (developmental fate of cells)?

___A) Differentiation occurs before determination

___B) In animals, the cells that appear after the first two division are determined

___C) A determined cell will keep its features wherever it is transported in the embryo

___D) When a cell is determined its structure (shape) will begin to change

___E) A determined cell has the same transcription model as a differentiated one

BEHAVIOUR

116. In the picture, a flying dummy (silhouette) of a bird is illustrated. If the dummy is moved over hatched chicks of a pheasant (*Phasianus colchicus*) from left to right (upper arrow) or from right to left (ie. moving backward) (lower arrow), the chicks will react as follows (mark the correct answer with an “X”):

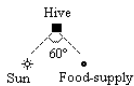
- I. In both cases the chicks will react to the silhouette by crouching**
 - II. In both cases the chicks will not react at all**
 - III. During the movement of the silhouette from left to right (upper arrow) the chicks will not react**
 - IV. During the movement of the silhouette from right to left (lower arrow) the chicks will not react**
 - V. During the movement of the silhouette from right to left (lower arrow) the chicks will react by crouching**
 - VI. During the movement of the silhouette from left to the right (upper arrow) the chicks react by crouching**
- ___A) Only I ___B) Only II ___C) III and V
- ___D) IV and VI ___E) III and VI

117. “Animal aggression comes out in several cases and aggression is also motivated by various conditions such as an external stimulus”.

Which of the following is not an aggressive behaviour?

- ___A) The behaviour of the prey that is under the threat of being killed
- ___B) Behaviour that does not reflect the normal behaviour and specifications of a group
- ___C) The behaviour against intruder in order to protect their territory
- ___D) Behaviour towards other animals that try to steal their own food
- ___E) Behaviour of a predator against its prey

118.

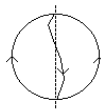


The location of the food-supply

Tail-wagging dance

When honeybees find a food-supply, they can show the exact place of the supply to other individuals of the colony by a “ tail-wagging dance”. An example of this behaviour is illustrated above

According to this example, for a honeybee that makes the illustrated tail-wagging dance below, which one of the following shows the location of the food-supply ?



___A)

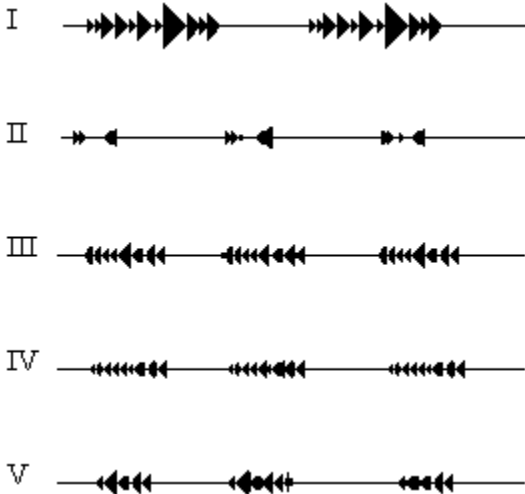
___B)

___C)

___D)

___E)

119.



It is known that some grasshopper species may make a sound in order to court the opposite sex and these sounds are species specific. It is also observed that these sounds seem very different when a close relative species live in the same area.

The sonograms of the “mating songs” of the five different species of male grasshoppers from the genus *Chorthippus* is given above. Which of the species (I-V) are living together in the same area?

- ___A) I and II
- ___B) III and IV
- ___C) III and V
- ___D) IV and V
- ___E) III, IV and V

120. “When a goose notices an egg outside her nest, she rises, extends her neck, touches the egg with her beak, and then rolls it back in very gently. She completes the same recovery behavior whether the objects she sees is a beer bottle or golf ball, even when the object is removed after she has begun to reach for it.”

Which of the following statements is correct according to the situation above?

- ☐ A) The reason why the goose rolls back the objects that do not structurally look like an egg is her lack of recognition
- ☐ B) This behaviour is caused by instinct
- ☐ C) In order to fill her nest to provide suitable incubation conditions
- ☐ D) This behavior is learnt from the parents.
- ☐ E) The goose recognises her own egg shape. For this reason, egg rolling is a fixed action pattern and continues without another stimulus

ECOLOGY

121. Deleted

122. Which one of the following environmental conditions affects the dispersal trend of a population positively?

- ___A) The conditions that cause high mortality sometimes create empty habitats
- ___B) Very frequent disturbances in the habitat conditions
- ___C) Absence of suitable habitats very close to each other
- ___D) A low level of natality causing the differences between the habitats
- ___E) A breakdown at any level of the food chain

123. Deleted

124. According to the population growth curves given below, which population has reached equilibrium by responding properly to negative feedback mechanisms with time?

- | | |
|---------------|---------|
| ___A) limit K | ___D) |
| | limit K |
| ___B) limit K | ___E) |
| | limit K |
| ___C) limit K | |

125. Deleted

126. A mosquito species which lives in hot and highly humid environment generally chooses little isolated aquatic habitats to reproduce and completes its larval development. This species gives many generations by reproducing in late May and early October. To increase its population in a given area which in the following is the most important limiting factor for this species which is very sensitive to the chemical changes in the habitat water .

- ☐ A) Increase in the saturation deficit in the air during the reproductive season
- ☐ B) Predation
- ☐ C) Competition with another species in the microhabitat
- ☐ D) Increase in the shadow factor
- ☐ E) Increase in relative humidity

127. Which one of the following explanations cannot be given about the relationship between the carrying capacity and the environmental response of populations with a high density?

- ☐ A) Competition increases
- ☐ B) The natality (birth) rate decreases
- ☐ C) The negative feedback mechanism works
- ☐ D) The environmental response decreases
- ☐ E) The mortality rate increases

128. Deleted

129. Deleted.

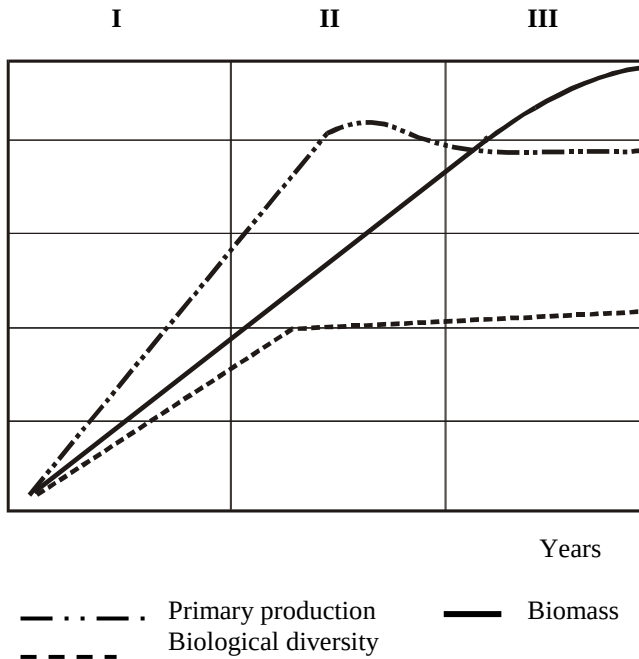
130. Deleted

131. Which one of the following cannot be said about the distribution of populations?

132. A park was built in a place that was occupied by a lot of species "A" trees. A long time ago they were all cut down no species "A" trees remained. Later gardeners planted more species "A" trees and also species "B" trees and species "C" trees had species B and C never grew in that region before. Nobody took care of this garden. After 100 years there were a lot of new "A" trees and "B" trees, but no young "C" trees . Which processes refer to the "A", "B" and "C" trees in that park?

<u>A</u>	<u>B</u>	<u>C</u>
___A) Introduced,	Acclimatized,	Reacclimatized
___B) Acclimatized,	Introduced,	Reacclimatized
___C) Introduced,	Reacclimatized,	Acclimatized
___D) Reacclimatized,	Acclimatized,	Introduced
___E) Acclimatized,	Reacclimatized,	Introduced

133. The graph represents the changes in the biomass, diversity and primary production in the ecological succession. Which blocks on the graph represent the first settler stages and the climax stage, respectively.



- ___A) I and II
___B) II and III
___C) I and III
___D) I, II and III
___E) None

134. An ecologist wants to investigate if there are any differences in the vegetation on the north and south facing sides of a valley. She lays down a rope from the top to the bottom of the slope and every 2 meters she places a 1 m² quadrat next to the rope. Standing above the quadrat she estimates and records the area occupied by each plant species. This technique involves which of the following?

- 1. The use of a point quadrat**
- 2. The recording of % cover**
- 3. The plotting of the results on a kite diagram**
- 4. Random sampling**
- 5. The use of a belt transect**

___A) 1 & 2

___B) 2 & 3

___C) 3, 4 & 5

___D) 1, 2 & 3

___E) 2, 3 & 5

135. Deleted

Answer Key-Theoretical Test-Part A

4. 4. Answer Key to the Theoretical Test

4.4.1. Part A

1. D	35. A	69. B	103. A
2. E	36. C	70. C	104. D
3. B	37. Deleted	71. D	105. A
4. C	38. B	72. E	106. A
5. Deleted	39. Deleted	73. E	107. Deleted
6. D	40. E	74. C	108. C
7. A	41. B	75. Deleted	109. E
8. D	42. A	76. D	110. A
9. B	43. C	77. B	111. Deleted
10. C	44. C	78. E	112. Deleted
11. B	45. A	79. D	113. Deleted
12. B	46. Deleted	80. E	114. B
13. C	47. B	81. Deleted	115. C
14. A	48. E	82. D	116. C
15. C	49. B	83. D	117. C
16. D	50. Deleted	84. Deleted	118. A
17. E	51. E	85. Deleted	119. A
18. C	52. A	86. E	120. E
19. B	53. A	87. Deleted	121. Deleted
20. D	54. Deleted	88. Deleted	122. A
21. C	55. C	89. D	123. Deleted
22. E	56. E	90. D	124. C
23. A	57. B	91. C	125. Deleted
24. E	58. E	92. B	126. A
25. B	59. Deleted	93. E	127. D
26. C	60. E	94. Deleted	128. Deleted
27. C	61. Deleted	95. Deleted	129. Deleted
28. E	62. D	96. C	130. Deleted
29. D	63. D	97. C	131. A
30. A	64. Deleted	98. A	132. D
31. D	65. Deleted	99. D	133. A
32. A	66. Deleted	100. C	134. E
33. D	67. Deleted	101. C	135. Deleted
34. D	68. E	102. Deleted	

Report of the 11th IBO in Antalya